

SmartContract Token Management System - Running Instruction

1.Pre-requisities Libraries

- Truffle
- Ganache
- Flask and Requests
- Web3

2. Go to project folder

1. Start Local Blockchain Command

```
Instructor@instructor-virtual-machine:~/Documents/Manoj/smartContract$ ganache-cli
ganache v7.9.2 (@ganache/cli: 0.10.2, @ganache/core: 0.10.2)
Starting RPC server
eth_getBlockByNumber
eth_getBlockByNumber

Available Accounts
=====
(0) 0xBd35847D0a1BA8b099005702230804F96936646F (1000 ETH)
(1) 0xC0d486B59296d90Ea6ee6329a6F7612d8f728B07 (1000 ETH)
(2) 0xF34E74Dc7Ec8471e3Ca71d47F0217251BDa4dD68 (1000 ETH)
(3) 0x18E4c852774Bff4b67d94Fc8e26921cF47f8B492 (1000 ETH)
(4) 0x261BB6dF346D20832107d3D8d7B048bF5ee6a68a (1000 ETH)
(5) 0x3c5aDc3BaCa678E2F0a84489f8e6E195beC67E49 (1000 ETH)
(6) 0x2AF5c587890819AdA7F7f69e64f3e49e0583b4bA (1000 ETH)
(7) 0x8018d54B6085910b4cA9C56A9B8A607870DAbD45 (1000 ETH)
(8) 0x7c271C004e00F1f6a05CEc8502f799Ba2d0C791E (1000 ETH)
(9) 0x42dF7c95b0164021770958fF74b93540125B1789 (1000 ETH)

Private Keys
=====
(0) 0x68af67b4d95aeac5d256a1525d6f333685b06b4c244bf6fe6aedef32c0f04a33
(1) 0xe36fae8c6c03082c1ca4c92b4aa12b6d22316b09d12e1067c4cc3b106b339e
(2) 0xf1f7dd890f8f10d02eba7bc8386ebb4048ac37c7fd7a61c1cff6d9a845d9bf94
(3) 0x337235b1da131178987f6ebdf1189ad20a469deee660c46c7d9c38cb2c34ffb0
(4) 0x3f76109a827352e2f896027d72e48724d0f18386aea224f3b5b57a74c58f80e6
(5) 0x3b9de393ca7f3437987a6090bce5bc9df1935a9dc585448925ceb8e1ecd8212a
(6) 0xaa57b6c70c73e52d3442d2b13f135bdf3a8e09440cbb5c0af561aacfd5c6a4c3
```

2. Compile Smart Contract Command

```

instructor@instructor-virtual-machine:~/Documents/Manoj/smartContract/contracts$ truffle compile

Compiling your contracts...
=====
> Compiling ./contracts/Migrations.sol
> Compiling ./contracts/TokenManager.sol
> Artifacts written to /home/instructor/Documents/Manoj/smartContract/build/contracts
> Compiled successfully using:
   - solc: 0.8.10+commit.fc410830.Emscripten.clang

```

3. Deploy Smart Contract

```

instructor@instructor-virtual-machine:~/Documents/Manoj/smartContract$ truffle migrate

Compiling your contracts...
=====
> Everything is up to date, there is nothing to compile.

Starting migrations...
=====
> Network name:      'development'
> Network id:        1772851552787
> Block gas limit: 30000000 (0x1c9c380)

1_initial_migration.js
=====

Deploying 'Migrations'
-----
> transaction hash: 0x03566359f393de34f687bf414c24b35dfbe16c3cfd740cccc0a858b7cea5c76f
> Blocks: 0        Seconds: 0
> contract address: 0xBfC903DAF8FF7478CA9924041348c1a3668f0a61
> block number:     1
> block timestamp:  1772852640
> account:          0x26c1352fc00143dEbe3045BD916cb7B3Ee236b85
> balance:          999.9991555345
> gas used:         250212 (0x3d164)
> gas price:        3.375 gwei
> value sent:       0 ETH
> total cost:       0.0008444655 ETH

```

4. Interact with Contract(Test)

```

instructor@instructor-virtual-machine:~/Documents/Manoj/smartContract/migrations$ truffle console
truffle(development)> let token = await TokenManager.deployed()
undefined

```

1. Deposit

```

truffle(development)> await token.Deposit("user1",100)
{
  tx: '0x14dcf1eef4bfc6143d9358fe16271933082f146a36ca2ab7eef3478b0f8ee309',
  receipt: {
    transactionHash: '0x14dcf1eef4bfc6143d9358fe16271933082f146a36ca2ab7eef3478b0f8ee309',
    transactionIndex: 0,
    blockNumber: 5,
    blockHash: '0x0698108aae1fed676408e29bbd7b4aa4f8b88b7d80992658422f43fb5da8ceef',
    from: '0x26c1352fc00143debe3045bd916cb7b3ee236b85',
    to: '0x4d7d6af07eb32dd4e7a7fde999ecb82e21e135b7',
    cumulativeGasUsed: 45582,

```

2. Balance Check

```
truffle(development)> let balance = await token.Query("user1")
balance.toString()undefined
truffle(development)> balance.toString()
'100'
```

3. Withdraw

```
truffle(development)> await token.Withdraw("user1",40)
{
  tx: '0xc2f5e9b351addf1fa2f3ba4a0550162997cd822913ddfba809aab04038650744',
  receipt: {
    transactionHash: '0xc2f5e9b351addf1fa2f3ba4a0550162997cd822913ddfba809aab04038650744',
    transactionIndex: 0,
    blockNumber: 6,
    blockHash: '0xf53e875614ee40734df40a3ee47f3adbdf84707cd5f06622bde687180675deb6',
    from: '0x26c1352fc00143debe3045bd916cb7b3ee236b85',
    to: '0x4d7d6af07eb32dd4e7a7fde999ecb82e21e135b7',
    cumulativeGasUsed: 29205,
```

4. Balance Check

```
truffle(development)> let newBalance = await token.Query("user1")
undefined
truffle(development)> newBalance.toString()
'60'
```

3. Change Contract Address in [middleware.py](#) file

4. Start Server

```
Instructor@instructor-virtual-machine:~/Documents/Manoj/smartContract/gateway$ python3 server.py
* Serving Flask app 'server'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI
server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8080
* Running on http://192.168.15.2:8080
Press CTRL+C to quit
127.0.0.1 - - [07/Mar/2026 13:21:48] "POST /deposit HTTP/1.1" 200 -
127.0.0.1 - - [07/Mar/2026 13:21:48] "GET /query/user1 HTTP/1.1" 200 -
127.0.0.1 - - [07/Mar/2026 13:21:48] "POST /withdraw HTTP/1.1" 200 -
127.0.0.1 - - [07/Mar/2026 13:21:48] "POST /set_policy HTTP/1.1" 200 -
127.0.0.1 - - [07/Mar/2026 13:21:48] "GET /get_policy/user1 HTTP/1.1" 200 -
```

4. Run Client

```
instructor@instructor-virtual-machine:~/Documents/Manoj/smartContract/gateway$ python3 client.py
{'result': 'Deposited 100 tokens to user1'}
{'balance': 170}
{'result': 'Withdrew 30 tokens from user1'}
{'balance': 140}
{'result': "Policy set for user1 as 'active'"}
{'policy': 'active'}
```

Project Structure

```
smartContract/
├── build/
│   ├── contracts/
│   │   ├── TokenManager.json
│   │   └── Migrations.json
│   ├── contracts/
│   │   ├── TokenManager.sol
│   │   └── Migrations.sol
│   ├── migrations/
│   │   ├── 1_initial_migration.js
│   │   └── 2_deploy_token.js
│   ├── middleware/
│   │   └── middleware.py
│   ├── gateway/
│   │   ├── server.py
│   │   └── client.py
│   ├── src/
│   │   └── requirements.py
│   └── test/
└── truffle-config.js
```